



Red Hat OpenShift AI: Supported Configurations for 3.x

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This article lists the Red Hat OpenShift AI (RHOAI) 3.x offering, the RHOAI components, their current support phase, and their compatibility with the underlying platforms.

⚠ ATTENTION: For Red Hat OpenShift AI 2.x go to the supported pages [HERE](#).

Red Hat OpenShift AI Self-Managed

You install **OpenShift AI Self-Managed** by installing the Red Hat OpenShift AI Operator and then configuring the Operator to manage standalone components of the product.

RHOAI Self-Managed is supported on any supported OpenShift Container Platform configuration running on x86_64, ppc64le, s390x and aarch64 architectures. This includes, but is not limited to, the following deployment topologies:

- Bare Metal
- Single Node OpenShift (SNO)
- Hosted control planes (on Bare Metal, OpenShift Virtualization, or cloud providers)

- IBM Cloud
- Red Hat OpenStack
- Amazon Web Services
- Google Cloud Platform
- Microsoft Azure
- VMware vSphere
- Nutanix
- Oracle Cloud
- IBM Power
- IBM Z

Note: RHOAI GPU workloads can be utilized on Hosted control planes with OpenShift Virtualization with GPU passthrough, as per the GPU provider's (e.g. NVIDIA) documentation for supported GPU models and feature availability, and subject to HCP and oVirt's GPU support.

This also includes support for RHOAI Self-Managed on managed OpenShift offerings such as OpenShift Dedicated, Red Hat OpenShift Service on AWS (ROSA with HCP), Red Hat OpenShift Service on AWS (classic architecture), Microsoft Azure Red Hat OpenShift, and OpenShift Kubernetes Engine. Currently, RHOAI Self-Managed is not supported on OpenShift running on platforms such as MicroShift.

For a full overview of the RHOAI Self-Managed life cycle and the currently supported releases, visit [this page](#).

Architecture, Version and Components

x86_64 architecture

RHOAI						
Operator Version		3.3		3.4		3.5
OpenShift Supported Versions		4.19.9+, 4.20, 4.21		4.19.9+, 4.20, 4.21, 4.22		4.19.9+, 4.20, 4.21, 4.22
Components	Status	Version	Status	Version	Status	Version
Dashboard	GA	2.0.0	GA	2.0.0	GA	2.0.0
AI Pipelines	GA	2.5.0	GA	2.16.0	GA	2.16.0
↳ Argo Workflows	GA	v3.6.7	GA	v3.7.3	GA	v3.7.3
Distributed Inference with llm-d - (1)	GA	0.4	GA	0.7.1	GA	0.9.0

RHOAI Operator Version		3.3		3.4		3.5
Feature Store (Feast)	GA	0.59.0	GA	0.62.0	GA	0.65.0
KServe	GA	0.15	GA	0.17.0	GA	0.19.0
Red Hat AI Inference (2)	-	-	GA	3.4.0	GA	3.5.0
Kubeflow Training Operator v1	Deprecated	1.9.0	Deprecated	1.9.0	Deprecated (3)	1.9.0
Kubeflow Trainer v2	GA	2.1.0	GA	2.1.0	GA	2.1.0
KubeRay	GA	1.4.2	GA	1.4.2	GA	1.4.2
Llama Stack Operator	TP	0.6.0	TP	0.9.0	-	-
OGX Operator	-	-	-	-	GA	0.11.0
↳ OGX	-	-	-	-	GA	1.1.2
Model as a Service (MaaS)	TP	0.0.2	GA	0.1.1	GA	0.2.0
MLflow	TP	3.6.0	GA	3.10.1	GA	3.14.0
MLServer	GA	1.7.1	GA	1.7.1	GA	1.7.1
Spark Operator	-	-	TP	2.4.0	GA	2.4.0
↳ Spark	-	-	TP	4.0.1	GA	4.0.1
TrustyAI	GA	1.37.0	GA	1.37.0	GA	1.37.0
↳ LM-Eval	GA	0.4.8	GA	0.4.8	GA	0.4.8
↳ NeMo Guardrails	TP	0.9.4	GA	0.9.4	GA	0.9.4
AI Hub	GA	0.3.5	GA	0.3.9	GA	0.3.10
Kubeflow Notebook Controller	GA	1.10.0	GA	1.10.0	GA	1.10.0
AutoML	-	-	TP	1.11.0	TP	1.11.0

RHOAI Operator Version		3.3		3.4		3.5
AutoRAG	-	-	TP	1.11.0	TP	1.11.0
EvalHub	-	-	-	-	TP	0.3.0

ARM (aarch64) architecture

RHOAI Operator Version		3.3		3.4		3.5
OpenShift Supported Versions		4.19.9+, 4.20, 4.21		4.19.9+, 4.20, 4.21, 4.22		4.19.9+, 4.20, 4.21, 4.22
Components	Status	Version	Status	Version	Status	Version
Dashboard	GA	2.0.0	GA	2.0.0	GA	2.0.0
AI Pipelines	GA	2.5.0	GA	2.16.0	GA	2.16.0
↳ Argo Workflows	GA	v3.6.7	GA	v3.7.3	GA	v3.7.3
Distributed Inference with IIm-d - (1)	GA	0.4	GA	0.7.1	GA	0.9.0
Feature Store (Feast)	GA	0.59.0	GA	0.62.0	GA	0.65.0
KServe	GA	0.15	GA	0.17.0	GA	0.19.0
Red Hat AI Inference (2)	-	-	GA	3.4.0	GA	3.5.0
Kubeflow Training Operator v1	Deprecated	1.9.0	Deprecated	1.9.0	Deprecated (3)	1.9.0
Kubeflow Trainer v2	GA	2.1.0	GA	2.1.0	GA	2.1.0

RHOAI Operator Version		3.3		3.4		3.5
KubeRay	GA	1.4.2	GA	1.4.2	GA	1.4.2
Llama Stack Operator	TP	0.6.0	TP	0.9.0	-	-
OGX Operator	-	-	-	-	GA	0.11.0
↳ OGX	-	-	-	-	GA	1.1.2
Model as a Service (MaaS)	TP	0.0.2	GA	0.11	GA	0.2.0
MLflow	-	-	-	-	-	-
MLServer	-	-	-	-	-	-
Spark Operator	-	-	-	-	-	-
↳ Spark	-	-	-	-	-	-
TrustyAI	GA	1.37.0	GA	1.37.0	GA	1.37.0
↳ LM-Eval	GA	0.4.8	GA	0.4.8	GA	0.4.8
↳ NeMo Guardrails	-	-	-	-	-	-
AI Hub	GA	0.3.5	GA	0.3.9	GA	0.3.10
Kubeflow Notebook Controller	GA	1.10.0	GA	1.10.0	GA	1.10.0
AutoML	-	-	-	-	-	-
AutoRAG	-	-	-	-	-	-
EvalHub	-	-	-	-	-	-

IBM Power (ppc64le) architecture

RHOAI Operator Version		3.3		3.4		3.5
OpenShift Supported Versions		4.19.9+, 4.20, 4.21		4.19.9+, 4.20, 4.21, 4.22		4.19.9+, 4.20, 4.21, 4.22
Components	Status	Version	Status	Version	Status	Version
Dashboard	GA	2.0.0	GA	2.0.0	GA	2.0.0
AI Pipelines	GA	2.5.0	GA	2.16.0	GA	2.16.0
↳ Argo Workflows	GA	v3.6.7	GA	v3.7.3	GA	v3.7.3
Distributed Inference with llm-d	-	-	-	-	-	-
Feature Store (Feast)	TP	0.59.0	TP	0.62.0	TP	0.65.0
KServe	GA	0.15	GA	0.17.0	GA	0.19.0
Red Hat AI Inference (2)	-	-	GA	3.4.0	GA	3.5.0
Kubeflow Training Operator v1	Deprecated	1.9.0	Deprecated	1.9.0	Deprecated (3)	1.9.0
Kubeflow Trainer v2	-	-	-	-	-	-
KubeRay	-	-	TP	1.4.2	TP	1.4.2
Llama Stack Operator	-	-	TP	0.9.0	-	-
OGX Operator	-	-	-	-	GA	0.11.0
↳ OGX	-	-	-	-	GA	1.1.2
Model as a Service (MaaS)	-	-	-	-	-	-
MLflow	-	-	-	-	TP	3.14.0
MLServer	-	-	-	-	TP	1.7.1

RHOAI Operator Version		3.3		3.4		3.5
Spark Operator	-	-	-	-	TP	2.4.0
↳ Spark	-	-	-	-	TP	4.0.1
TrustyAI	GA	1.37.0	GA	1.37.0	GA	1.37.0
↳ LM-Eval	GA	0.4.8	GA	0.4.8	GA	0.4.8
↳ NeMo Guardrails	TP	0.9.4	GA	0.9.4	GA	0.9.4
AI Hub	GA	0.3.5	GA	0.3.9	GA	0.3.10
Kubeflow Notebook Controller	GA	1.10.0	GA	1.10.0	GA	1.10.0
AutoML	-	-	TP	1.11.0	TP	1.11.0
AutoRAG	-	-	-	-	-	-
EvalHub	-	-	-	-	-	-

IBM Z (s390x) architecture

RHOAI Operator Version		3.3		3.4		3.5
OpenShift Supported Versions		4.19.9+, 4.20, 4.21		4.19.9+, 4.20, 4.21, 4.22		4.19.9+, 4.20, 4.21, 4.22
Components	Status	Version	Status	Version	Status	Version
Dashboard	GA	2.0.0	GA	2.0.0	GA	2.0.0
AI Pipelines	-	-	-	-	-	-
↳ Argo Workflows	-	-	-	-	-	-
Distributed Inference with IIm-d	-	-	-	-	-	-

RHOAI Operator Version		3.3		3.4		3.5
Feature Store (Feast)	-	-	-	-	-	-
KServe	GA	0.15	GA	0.17.0	GA	0.19.0
Red Hat AI Inference (2)	-	-	GA	3.4.0	GA	3.5.0
Kubeflow Training Operator v1	-	-	-	-	-	-
Kubeflow Trainer v2	-	-	-	-	-	-
KubeRay	-	-	-	-	-	-
Llama Stack Operator	-	-	-	-	-	-
OGX Operator	-	-	-	-	-	-
↳ OGX	-	-	-	-	-	-
Model as a Service (MaaS)	-	-	-	-	-	-
MLflow	-	-	-	-	-	-
MLServer	-	-	-	-	-	-
Spark Operator	-	-	-	-	-	-
↳ Spark	-	-	-	-	-	-
TrustyAI	GA	1.37.0	GA	1.37.0	GA	1.37.0
↳ LM-Eval	-	-	-	-	-	-
↳ NeMo Guardrails	-	-	-	-	-	-
IBM Spyre Operator	GA	1.1.1	GA	1.2.1	GA	TBD
AI Hub	GA	0.3.5	GA	0.3.9	GA	0.3.10
Kubeflow Notebook Controller	GA	1.10.0	GA	1.10.0	GA	1.10.0
AutoML	-	-	-	-	-	-
AutoRAG	-	-	-	-	-	-
EvalHub	-	-	-	-	-	-

Notes

(1) Distributed Inference with llm-d requires OpenShift 4.20 or later

(2) For supported AI accelerators for Red Hat AI Inference Server, see Supported AI accelerators for Red Hat AI Inference Server

(3) Kubeflow Training Operator v1 is supported through the Full Support phase of Red Hat OpenShift AI 3.5 only. It is not supported during the Extended Update Support (EUS) phase of 3.5. Support for this component ends when 3.5 exits Full Support (**March 27, 2027**). After that date, Kubeflow Training Operator v1 is End of Life (EOL). Customers should migrate to Kubeflow Trainer v2 before that date.

TP: Technology Preview

DP: Developer Preview

Developer and Technology Previews: How they compare

LA: Limited Availability. During this phase you can install and receive support for the feature only with specific approval from Red Hat. Without such approval, the feature is unsupported.

GA: General Availability.

EUS: Extended Update Support. During the EUS phase, Red Hat will maintain component specific support.

EOL: End of Life. During this phase, the component will no longer be supported.

Deprecated (EUS exception): The component remains Deprecated and supported through Full Support of the listed release, but is excluded from Extended Update Support. See the component footnote for the support end date.

RHOAI and vLLM version compatibility

vLLM runtime versions are maintained as part of Red Hat AI Inference (RHAI). For the vLLM version compatibility matrix, supported accelerators, and related configuration details, see Red Hat AI Inference - Supported Product and Hardware Configurations.

Red Hat OpenShift AI Operator Dependencies

For information on the compatibility and supported versions of Red Hat OpenShift AI Operator dependencies, see the following documentation:

- Red Hat OpenShift Serverless: release notes
- Red Hat OpenShift Service Mesh: release notes
- Red Hat Build of Kueue: release notes
- Node Feature Discovery Operator: documentation
- Red Hat - Authorino Operator: documentation
- NVIDIA GPU Operator: documentation

- Intel Gaudi Base Operator: [documentation](#)
- AMD GPU Operator: [documentation](#)
- NVIDIA Network Operator: [documentation](#)
- Red Hat Connectivity Link: [release notes](#)
- IBM Spyre Operator : [documentation](#)
- cert-manager Operator for Red Hat OpenShift: [documentation](#)
- PostgreSQL Operator: required for Llama Stack server deployments (3.4+)
- Red Hat Leader Worker Set Operator: required for Distributed Inference with Llm-d

Starting with RHOAI 3.4, Red Hat Connectivity Link (RHCL) is included in Red Hat AI SKUs for Models as a Service (MaaS) use cases only. For all other RHOAI components, RHCL is not included. When MaaS is enabled, RHCL is deployed via the Kuadrant Operator, which provides authentication, authorization, and rate limiting. For the RHCL version compatibility matrix, see [Red Hat Connectivity Link Supported Configurations](#).

Red Hat OpenShift AI does not directly support any specific accelerators. To use accelerator functionality in OpenShift AI, the relevant accelerator Operators are required. OpenShift AI supports integration with the relevant Operators, and provides many images across the product that include the libraries to work with NVIDIA GPUs, AMD GPUs, Intel Gaudi AI accelerators and IBM Spyre. For more information about which devices are supported by an Operator, see the [documentation](#) for that Operator.

Support requirements and limitations

Review this section to understand the requirements for Red Hat support and any limitations to Red Hat support of Red Hat OpenShift AI.

Supported browsers

- Google Chrome
- Mozilla Firefox
- Safari

Supported services

Red Hat OpenShift AI supports the following services:

Service Name	Description
EDB Postgres AI - solution including Pgvector	Use powerful hybrid search for AI RAG and multimodal AI recommender applications with EDB's vector database solution including Pgvector. Combine AI, transactional, and analytical workloads with native vector index search, enterprise-grade security, and scalability in a unified Postgres environment.
Elasticsearch	Build transformative RAG applications, proactively resolve observability issues, and address complex security threats – all with the power of Search AI.

Service Name	Description
IBM Watson Studio	IBM® watsonx.ai is part of the IBM watsonx AI and data platform, bringing together new generative AI capabilities powered by foundation models and traditional machine learning (ML) into a powerful studio spanning the AI lifecycle.
Intel® oneAPI AI Analytics Toolkit Container	The AI Kit is a set of AI software tools to accelerate end-to-end data science and analytics pipelines on Intel® architectures.
NVIDIA NIM	NVIDIA NIM, part of NVIDIA AI Enterprise, is a set of easy-to-use microservices designed for secure, reliable deployment of high-performance AI model inferencing across the cloud, data center and workstations. Supporting a wide range of AI models, including open-source community and NVIDIA AI Foundation models, it ensures seamless, scalable AI inferencing, on-premises or in the cloud, leveraging industry standard APIs.
OpenVINO	OpenVINO is an open source toolkit to help optimize deep learning performance and deploy using an inference engine onto Intel® hardware.
Pachyderm	Pachyderm is the data foundation for machine learning. It provides industry-leading pipelines, data versioning, and lineage for data science teams to automate the machine learning lifecycle.
Starburst Enterprise	Starburst Enterprise platform (SEP) is the commercial distribution of Trino, which is an open-source, Massively Parallel Processing (MPP) ANSI SQL query engine. Starburst simplifies data access for your Red Hat OpenShift AI workloads by providing fast access to all of your data, no matter where it lives. Starburst does this by connecting directly to each data source and pulling the data back into memory for processing, alleviating the need to copy or move the data into a single location first.
Jupyter	Jupyter is a multi-user version of the notebook designed for companies, classrooms, and research labs.

Supported workbench images

The latest supported workbench images in Red Hat OpenShift AI are installed with Python by default.

You can install packages that are compatible with the supported version of Python on any workbench server that has the binaries required by that package. If the required binaries are not included on the workbench image you want to use, contact Red Hat Support to request that the binary be considered for inclusion.

To provide a consistent, stable platform for your model development, select workbench images that contain the same version of Python. Workbench images available on OpenShift AI are pre-built and ready for you to use immediately after OpenShift AI is installed or upgraded.

Workbench images are supported for a minimum of one year. Major updates to pre-configured workbench images occur about every six months. Therefore, two supported workbench image versions are typically available at any given time. You can use this support period to update your code to use components from the latest available workbench image. Legacy workbench image versions, that is, not the two most recent versions, might still be available for selection. Legacy image versions include a label that indicates the image is out-of-date. To use the latest package versions, Red Hat recommends that you use the most recently added workbench image. If necessary, you can still access older workbench images from the registry, even if they are no longer supported. You can then add the older workbench images as custom workbench images to cater for your project’s specific requirements.

Workbench images denoted with *Technology Preview* in the following table are not supported with Red Hat production service level agreements (SLAs) and might not be functionally complete. Red Hat does not recommend using Technology Preview features in production. These features provide early access to upcoming product features, enabling customers to test functionality and provide feedback during the development process.

Notebooks Supported on x86_64

Image name	Image version	Preinstalled packages
Code Server Data Science CPU Python 3.12	2025.2 (Recommended)	code-server: 4.104, Python 3.12, Boto3: 1.40, Kafka-Python-ng: 2.2, Matplotlib: 3.10, NumPy: 2.3, Pandas: 2.3, scikit-learn: 1.7, Scipy: 1.16, Sklearn-onnx: 1.19, ipykernel: 6.30, Kubeflow-Training: 1.9
Jupyter Data Science CPU Python 3.12	2025.2 (Recommended)	Python 3.12, JupyterLab: 4.4, Boto3: 1.40, Kafka-Python-ng: 2.2, Kubeflow SDK: 2.14, Matplotlib: 3.10, NumPy: 2.3, Pandas: 2.3, scikit-learn: 1.7, Scipy: 1.16, Odh-Elyra: 4.3, PyMongo: 4.15, Pyodbc: 5.2, Codeflare-SDK: 0.34, Feast: 0.59, Sklearn-onnx: 1.19, Psycopg: 3.2, MySQL Connector/Python: 9.4, Kubeflow-Training: 1.9
Jupyter Minimal CPU Python 3.12	2025.2 (Recommended)	Python 3.12, JupyterLab: 4.4
Jupyter Minimal CUDA Python 3.12	2025.2 (Recommended)	CUDA 12.9, Python 3.12, JupyterLab: 4.4
Jupyter Minimal ROCm Python 3.12	2025.2 (Recommended)	ROCm 6.4, Python 3.12, JupyterLab: 4.4
Jupyter PyTorch LLM Compressor CUDA Python 3.12	2025.2 (Recommended)	CUDA 12.9, Python 3.12, PyTorch: 2.7, LLM-Compressor: 0.9, JupyterLab: 4.4, Tensorboard: 2.20, Boto3: 1.40, Kafka-Python-ng: 2.2, Kubeflow SDK: 2.14, Matplotlib: 3.10, NumPy: 2.3, Pandas: 2.3, scikit-learn: 1.7, Scipy: 1.16, Odh-Elyra: 4.3, PyMongo: 4.15, Pyodbc: 5.2, Feast: 0.59, Sklearn-onnx: 1.19, Psycopg: 3.2, MySQL Connector/Python: 9.4, Kubeflow-Training: 1.9

Image name	Image version	Preinstalled packages
Jupyter PyTorch CUDA Python 3.12	2025.2 (Recommended)	CUDA 12.9, Python 3.12, PyTorch: 2.7, JupyterLab: 4.4, Tensorboard: 2.20, Boto3: 1.40, Kafka-Python-ng: 2.2, Kubeflow SDK: 2.14, Matplotlib: 3.10, NumPy: 2.3, Pandas: 2.3, scikit-learn: 1.7, Scipy: 1.16, Odh-Elyra: 4.3, PyMongo: 4.15, Pyodbc: 5.2, Codeflare-SDK: 0.34, Feast: 0.59, Sklearn-onnx: 1.19, Psycopg: 3.2, MySQL Connector/Python: 9.4, Kubeflow-Training: 1.9
Jupyter PyTorch ROCm Python 3.12	2025.2 (Recommended)	ROCm 6.4, Python 3.12, ROCm-PyTorch: 2.7, JupyterLab: 4.4, Tensorboard: 2.20, Kafka-Python-ng: 2.2, Matplotlib: 3.10, NumPy: 2.3, Pandas: 2.3, scikit-learn: 1.7, Scipy: 1.16, Odh-Elyra: 4.3, PyMongo: 4.15, Pyodbc: 5.2, Codeflare- SDK: 0.34, Feast: 0.59, Sklearn-onnx: 1.19, Psycopg: 3.2, MySQL Connector/Python: 9.4, Kubeflow-Training: 1.9
Jupyter TensorFlow CUDA Python 3.12	2025.2 (Recommended)	CUDA 12.9, Python 3.12, TensorFlow: 2.20, JupyterLab: 4.4, Tensorboard: 2.20, Nvidia-CUDA-CU12-Bundle: 12.9, Boto3: 1.40, Kafka-Python-ng: 2.2, Kubeflow SDK: 2.14, Matplotlib: 3.10, NumPy: 2.1, Pandas: 2.3, scikit-learn: 1.7, Scipy: 1.16, Odh-Elyra: 4.3, PyMongo: 4.15, Pyodbc: 5.2, Codeflare-SDK: 0.34, Feast: 0.59, Sklearn-onnx: 1.19, Psycopg: 3.2, MySQL Connector/Python: 9.4
Jupyter TensorFlow ROCm Python 3.12	2025.2 (Recommended)	ROCm 6.4, Python 3.12, TensorFlow-ROCm: 2.18, JupyterLab: 4.4, Tensorboard: 2.18, Kafka-Python-ng: 2.2, Matplotlib: 3.10, NumPy: 1.26, Pandas: 2.3, scikit-learn: 1.7, Scipy: 1.16, Odh-Elyra: 4.3, PyMongo: 4.15, Pyodbc: 5.2, Codeflare-SDK: 0.34, Sklearn-onnx: 1.19, Psycopg: 3.2, MySQL Connector/Python: 9.4
Jupyter TrustyAI CPU Python 3.12	2025.2 (Recommended)	Python 3.12, JupyterLab: 4.4, TrustyAI: 0.6, Transformers: 4.56, Datasets: 4.0, Accelerate: 1.10, Torch: 2.7, Boto3: 1.40, Kafka-Python-ng: 2.2, Kubeflow SDK: 2.14, Matplotlib: 3.10, NumPy: 1.26, Pandas: 1.5, scikit-learn: 1.7, Scipy: 1.16, Odh-Elyra: 4.3, PyMongo: 4.15, Pyodbc: 5.2, Codeflare-SDK: 0.34, Sklearn-onnx: 1.19, Psycopg: 3.2, MySQL Connector/Python: 9.4, Kubeflow-Training: 1.9

Notebooks supported on IBM Power

Image name	Image version	Preinstalled packages
Code Server Data Science CPU Python 3.12	2025.2	code-server 1.104, Python 3.12, Boto3: 1.40, Kafka-Python-ng: 2.2, Matplotlib: 3.10, Numpy: 2.3, Pandas: 2.3, Scikit-learn: 1.7, Scipy: 1.16, Sklearn-onnx: 1.19, ipykernel: 6.30, Kubeflow-Training: 1.9
Jupyter Data Science CPU Python 3.12	2025.2	Python 3.12, JupyterLab: 4.4, Boto3: 1.40, Kafka-Python-ng: 2.2, Kfp: 2.14, Matplotlib: 3.10, Numpy: 2.3, Pandas: 2.3, Scikit-learn: 1.7, Scipy: 1.16, Odh-Elyra: 4.2, PyMongo: 4.14, Pyodbc: 5.2, Sklearn-onnx: 1.19, Psycopg: 3.2, MySQL Connector/Python: 9.3, Kubeflow-Training: 1.9
Jupyter Minimal CPU Python 3.12	2025.2	Python 3.12, JupyterLab: 4.4
Jupyter TrustyAI CPU Python 3.12	2025.2	Python 3.12, JupyterLab: 4.4, TrustyAI: 0.6, Transformers: 4.56, Datasets: 4.0, Accelerate: 1.10, Torch: 2.7, Boto3: 1.40, Kafka-Python-ng: 2.2, Kfp: 2.14, Matplotlib: 3.10, Numpy: 1.26, Pandas: 1.5, Scikit-learn: 1.7, Scipy: 1.16, Odh-Elyra: 4.2, PyMongo: 4.14, Pyodbc: 5.2, Sklearn-onnx: 1.19, Psycopg: 3.2, Kubeflow-Training: 1.9
Jupyter TrustyAI Minimal CPU Python 3.12	2025.2	IPython: 9.8.0, ipykernel: 7.1.0, ipywidgets: 8.1.2, jupyter_client: 8.7.0, jupyter_core: 5.9.1, jupyter_server: 2.17.0, jupyterlab: 4.4.9, nbclient: 0.10.2, nbconvert: 7.16.6, nbformat: 5.10.4, notebook: not installed, qtconsole: not installed, traitlets: 5.14.3

Notebooks Supported on IBM Z

Image name	Image version	Preinstalled packages
Code Server Data Science CPU Python 3.12	2025.2 (Recommended)	code-server 1.104, Python 3.12, Boto3: 1.40, Kafka-Python-ng: 2.2, Numpy: 2.3, Pandas: 2.3, Scikit-learn: 1.7, Scipy: 1.16, ipykernel: 6.30
Jupyter Data Science CPU Python 3.12	2025.2 (Recommended)	Python 3.12, JupyterLab: 4.4, Boto3: 1.40, Kafka-Python-ng: 2.2, Kfp: 2.14, Matplotlib: 3.10, Numpy: 2.3, Pandas: 2.3, Scikit-learn: 1.7, Scipy: 1.16, PyMongo: 4.14, Pyodbc: 5.2, Sklearn-onnx: 1.19, Psycopg: 3.2, MySQL Connector/Python: 9.3
Jupyter Minimal CPU Python 3.12	2025.2 (Recommended)	Python 3.12, JupyterLab: 4.4
	2024.2	Python 3.11, JupyterLab: 4.4

Image name	Image version	Preinstalled packages
Jupyter TrustyAI CPU Python 3.12	2025.2 (Recommended)	Python 3.12, JupyterLab: 4.4, TrustyAI: 0.6, Transformers: 4.56, Datasets: 4.0, Accelerate: 1.10, PyArrow 20.0, Torch: 2.7, Boto3: 1.40, Kafka-Python-ng: 2.2, Kfp: 2.14, Matplotlib: 3.10, Numpy: 1.26, Pandas: 1.5, Scikit-learn: 1.7, Scipy: 1.16, PyMongo: 4.14, Pyodbc: 5.2, Psycopg: 3.2

Supported model-serving runtimes

Runtime name	Description	Exported model format
vLLM Spyre AI Accelerator ServingRuntime for KServe	A high-throughput and memory-efficient inference and serving runtime that supports IBM Spyre AI accelerators on x86	Supported models
Caikit Text Generation Inference Server (Caikit-TGIS) ServingRuntime for KServe (1)	A composite runtime for serving models in the Caikit format	Caikit Text Generation
Caikit Standalone ServingRuntime for KServe (2)	A runtime for serving models in the Caikit embeddings format for embeddings tasks	Caikit Embeddings
OpenVINO Model Server	A scalable, high-performance runtime for serving models that are optimized for Intel architectures	PyTorch, TensorFlow, OpenVINO IR, PaddlePaddle, MXNet, Caffe, Kaldi
[Deprecated] Text Generation Inference Server (TGIS) Standalone ServingRuntime for KServe (3)	A runtime for serving TGI-enabled models	PyTorch Model Formats
vLLM NVIDIA GPU ServingRuntime for KServe	A high-throughput and memory-efficient inference and serving runtime for large language models that supports NVIDIA GPU accelerators	Supported models
vLLM Intel Gaudi Accelerator ServingRuntime for KServe	A high-throughput and memory-efficient inference and serving runtime that supports Intel Gaudi accelerators	Supported models

Runtime name	Description	Exported model format
vLLM AMD GPU ServingRuntime for KServe	A high-throughput and memory-efficient inference and serving runtime that supports AMD GPU accelerators	Supported models
vLLM CPU ServingRuntime for KServe	A high-throughput and memory-efficient inference and serving runtime that supports IBM Power (ppc64le) and IBM Z (s390x)	Supported models
vLLM Spyre s390x ServingRuntime for KServe	A high-throughput and memory-efficient inference and serving runtime that supports IBM Spyre accelerators on s390x (IBM Z)	Supported models
vLLM Spyre ppc64le ServingRuntime for KServe	A high-throughput and memory-efficient inference and serving runtime that supports IBM Spyre accelerators on ppc64le (IBM Power)	Supported models
MLServer ServingRuntime for KServe	A runtime designed to simplify the deployment of machine learning models	Scikit-Learn (sklearn), XGBoost, LightGBM, ONNX
MLServer NVIDIA GPU ServingRuntime for KServe	A GPU-accelerated inference and serving runtime for ONNX models that supports NVIDIA GPU accelerators	ONNX
AutoGluon (Technology Preview)	A runtime that enables streamlined deployment of AutoML models	AutoGluon TabularPredictor

(1) The composite Caikit-TGIS runtime is based on Caikit and Text Generation Inference Server (TGIS). To use this runtime, you must convert your models to Caikit format. For an example, see [Converting Hugging Face Hub models to Caikit format in the caikit-tgis-serving repository](#).

(2) The Caikit Standalone runtime is based on Caikit NLP. To use this runtime, you must convert your models to the Caikit embeddings format. For an example, see [Tests for text embedding module](#).

(3) The *Text Generation Inference Server (TGIS) Standalone ServingRuntime for KServe* is deprecated. For more information, see [Red Hat OpenShift AI release notes](#).

Deployment requirements for supported model-serving runtimes

Runtime name	Default protocol	Additional protocol	Single node OpenShift support	Deployment mode
vLLM Spyre AI Accelerator ServingRuntime for KServe	REST	No	Yes	Raw
Caikit Text Generation Inference Server (Caikit-TGIS) ServingRuntime for KServe	REST	gRPC	Yes	Raw
Caikit Standalone ServingRuntime for KServe	REST	gRPC	Yes	Raw
OpenVINO Model Server	REST	None	Yes	Raw
[Deprecated] Text Generation Inference Server (TGIS) Standalone ServingRuntime for KServe	gRPC	None	Yes	Raw
vLLM NVIDIA GPU ServingRuntime for KServe	REST	None	Yes	Raw
vLLM Intel Gaudi Accelerator ServingRuntime for KServe	REST	None	Yes	Raw
vLLM AMD GPU ServingRuntime for KServe	REST	None	Yes	Raw
vLLM CPU ServingRuntime for KServe (1)	REST	None	Yes	Raw
MLServer ServingRuntime for KServe	REST	No	Yes	Raw
MLServer NVIDIA GPU ServingRuntime for KServe	REST	No	Yes	Raw
AutoGluon (Technology Preview)	REST	No	Yes	Raw

(1) For vLLM CPU ServingRuntime for KServe, if you are using IBM Z and IBM Power architecture, you can only deploy models in standard deployment mode.

Tested and verified model-serving runtimes

Name	Description	Exported model format
NVIDIA Triton Inference Server	An open-source inference-serving software for fast and scalable AI in applications.	TensorRT, TensorFlow, PyTorch, ONNX, OpenVINO, Python, RAPIDS FIL, and more.
IBM Power accelerated Triton Inference Server	An open-source inference-serving software for fast and scalable AI in applications.	PyTorch, ONNX, Python and ML
IBM Z Accelerated for NVIDIA Triton Inference Server	An open-source AI inference server that standardizes model deployment and execution, delivering streamlined, high-performance inference at scale.	ONNX-MLIR, Snap ML (C++), PyTorch.

Deployment requirements for tested and verified model-serving runtimes

Name	Default protocol	Additional protocol	Single node OpenShift support	Deployment mode
NVIDIA Triton Inference Server	gRPC	REST	Yes	Standard (Raw)
IBM Power accelerated Triton Inference Server	REST	None	Yes	Standard (Raw)
IBM Z Accelerated for NVIDIA Triton Inference Server	gRPC	REST	Yes	Standard (Raw)

Training images

To run distributed training jobs in OpenShift AI, you can use one of the following types of training image:

- **Ray-based training images** – purpose-built images tested and verified for distributed training with Ray. See the table below.
- **Workbench images with the Kubeflow Training Operator (KFTO)** – the same x86_64 workbench images listed in the Supported workbench images section can also be used as base images for distributed training jobs with the Kubeflow Training Operator. You can install additional packages to create custom training images, as described in the product documentation.

Ray-based training images

The following table provides information about the latest available Ray-based training images in Red Hat OpenShift AI. These images are AMD64 images, which might not work on other architectures.

You can use the provided images as base images, and install additional packages to create custom images, as described in the product documentation. If the required packages are not included in the training image you want to use, contact Red Hat Support to request that the package be considered for inclusion, the images are Tested & Verified but not yet officially supported.

The images are updated periodically with new versions of the installed packages. These images have been tested and verified for the use cases and configurations that are documented in the corresponding product documentation. Bug fixes and CVE fixes are delivered after they are available in upstream packages, in newer versions of these images only; fixes are not backported to earlier image versions.

Image type	RHOAI version	Image version	URL	Preinstalled packages
CUDA	3.3	2.52.1-py312-cu128	quay.io/modh/ray:2.52.1-py312-cu128	Ray 2.52.1, CUDA 12.8, Python 3.12
	3.4	2.53.0-py312-cu128	quay.io/modh/ray:2.53.0-py312-cu128	Ray 2.53.0, CUDA 12.8, Python 3.12
	3.5	2.55.1-py312-cu128	quay.io/modh/ray:2.55.1-py312-cu128	Ray 2.55.1, CUDA 12.8, Python 3.12
Ray ROCm	3.3	2.52.1-py312-rocm62	quay.io/modh/ray:2.52.1-py312-rocm62	Ray 2.52.1, ROCm 6.2, Python 3.12
	3.4	2.53.0-py312-rocm64	quay.io/modh/ray:2.53.0-py312-rocm64	Ray 2.53.0, ROCm 6.4, Python 3.12
	3.5	2.55.1-py312-rocm64	quay.io/modh/ray:2.55.1-py312-rocm64	Ray 2.55.1, ROCm 6.4, Python 3.12

Kubeflow Trainer v2 ClusterTrainingRuntimes

The following table lists the pre-installed ClusterTrainingRuntimes available in RHOAI 3.4 for use with the Kubeflow Trainer v2 (`TrainJob` API). These runtimes provide ready-to-use templates for distributed PyTorch training jobs.

Runtime name	Framework	Default image	Accelerator	Preinstalled packages
training-hub	training-hub	registry.redhat.io/rhoai/odh-th06-cuda130-torch210-py312-rhel9	CUDA	Python 3.12, PyTorch: 2.0.1, Transformers: 4.34.0, Datasets: 2.12.0, Accelebrate: 1.12.0, PEFT: 0.11.0, TFTrainer: 0.24.0, DeepSpeed: 0.12.10, Unsloth: 2026.3.3, FlashAttention: 2.8.3, MLflow: 2.12.0, Kubeflow SDK: 0.3.0, Kserve: 0.14.0, Seldon: 2.4.4, Pandas: 2.3.3, sklearn: 1.5.2
training-hub-cpu	training-hub	registry.redhat.io/rhoai/odh-th06-cpu-torch210-py312-rhel9	CPU	Python 3.12, PyTorch: 2.0.1, Transformers: 4.34.0, Datasets: 2.12.0, Accelebrate: 1.12.0, PEFT: 0.11.0, TFTrainer: 0.24.0, MLflow: 2.12.0, Kubeflow SDK: 0.3.0, Kserve: 0.14.0, Seldon: 2.4.4, Pandas: 2.3.3, sklearn: 1.5.2
training-hub-rocm	training-hub	registry.redhat.io/rhoai/odh-th06-rocm64-torch210-py312-rhel9	ROCm	Python 3.12, PyTorch: 2.0.1, Transformers: 4.34.0, Datasets: 2.12.0, Accelebrate: 1.12.0, PEFT: 0.11.0, TFTrainer: 0.24.0, DeepSpeed: 0.12.10, Unsloth: 2026.3.3, FlashAttention: 2.8.3, MLflow: 2.12.0, Kubeflow SDK: 0.3.0, Kserve: 0.14.0, Seldon: 2.4.4, Pandas: 2.3.3, sklearn: 1.5.2
torch-distributed	torch	registry.redhat.io/rhoai/odh-th06-cuda130-torch210-py312-rhel9	CUDA	Python 3.12, PyTorch: 2.0.1, Transformers: 4.34.0, Datasets: 2.12.0, Accelebrate: 1.12.0, PEFT: 0.11.0, TFTrainer: 0.24.0, DeepSpeed: 0.12.10, Unsloth: 2026.3.3, FlashAttention: 2.8.3, MLflow: 2.12.0, Kubeflow SDK: 0.3.0, Kserve: 0.14.0, Seldon: 2.4.4, Pandas: 2.3.3, sklearn: 1.5.2

Runtime name	Framework	Default image	Accelerator	Preinstalled packages
torch-distributed-cpu	torch	registry.redhat.io/rhoai/odh-th06-cpu-torch210-py312-rhel9	CPU	Python 3.12, PyTorch: Transformers: 4.57.6, Datasets: 4.3.0, Accelerate: 1.12.0, PEFT: 0.18.1, Transformers: 0.24.0, MLflow: 3.10.1, Kubeflow SDK: 0.3.0, 2.4.4, Pandas: 2.3.3, sklearn: 1.8.0
torch-distributed-rocm	torch	registry.redhat.io/rhoai/odh-th06-rocm64-torch291-py312-rhel9	ROCm	Python 3.12, PyTorch: Transformers: 4.57.6, Datasets: 4.3.0, Accelerate: 1.12.0, PEFT: 0.18.1, Transformers: 0.24.0, DeepSpeed: 0.13.2, Unsloth: 2026.3.3, FlashAttention: 2.8.3, MLflow: 3.10.1, Kubeflow SDK: 0.3.0, 2.4.4, Pandas: 2.3.3, sklearn: 1.8.0

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